

## Semester Examination -2023

Course Code: PHYUGCC02

Course Title: Mechanics

Department of Appearing Students: PHYSICS, I SEM, 1<sup>st</sup> Year

Full Marks: 40

Time: 2 hrs

(Use separate answer sheet for each group)

Answer two questions from each group

Group-A

- (a) Calculate the gravitational potential and hence intensity at any point inside the solid sphere of radius R.

(b) Show that the centre of oscillation and centre of suspension for a given compound pendulum are interchangeable. What is the condition for minimum time period of a compound pendulum?

(c) Plot the variation of gravitational intensity and potential due to a hollow spherical shell with distance from its centre.

4+ (2+2)+2
- (a) Prove that the torsional rigidity of a cylindrical wire is  $G=\pi\eta r^4\phi/(2l)$ .

(b) What is flexural rigidity? A uniform beam is supported horizontally near the ends on two knife edges and a concentrated load W is applied at any point of the beam. Neglecting the weight of the beam, calculate the depression at any point of the beam.

(c) A steel rod of length 50cm, width 2cm and thickness 1cm is bent into the form of an arc of radius of curvature 2.0m. Calculate the bending moment. Young's modulus of the material of the rod= $2\times 10^{11}\text{N/m}^2$ .

4+4+2
- (a) What is ideal fluid? Deduce Poiseuille's formula for the flow of liquid through a narrow horizontal tube.

(b) Two vessels of equal cross section ' $\alpha$ ' are joining near their bases by a horizontal narrow tube of length L and internal radius ' $r$ '. Initially the liquid surfaces are at heights ' $5h$ ' and ' $h$ ' respectively above the capillary tube. Calculate the time taken for the difference in level become ' $h$ ' if the coefficient of viscosity is ' $\eta$ ' and density is ' $\rho$ '.

(c) The height of liquid inside a cylindrical vessel is H. Show that the range where the liquid strike the ground will be maximum when the hole is at a height H/2 from the bottom of the vessel.

(1+4)+3+2

